

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, VIII: THE TOWER-LEVEL HECKE–KIRCHBERG LIMIT AND THE NON-COMMUTATIVE IWASAWA K -MODULE

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ABSTRACT. The single open problem left by [VI] (Remark 4.3 there) asked whether the tail norms of [V, Thm. 3.1] lift from the sandpile groups to the shadow algebras $\mathcal{O}_{B(Y_k)}$ of the Iwahori tower, and whether the lifts assemble into one Kirchberg algebra whose K -theory carries the pro-module X_∞ of [V]. We answer both questions, and the answers differ from the expected shape in three places, each forced by a theorem. First, at the q -regular levels $k \geq 1$ the shadow must carry $q - 1$ return arcs, not q : the level-0 normalization of [VI] is the $(q+1)$ -regular special case, and with q return arcs the shadow K_0 is a finite group unrelated to $\text{Jac}(Y_k)$ and K_1 vanishes. Second, the tail map $\tau: Y_{k+1} \rightarrow Y_k$ extends to a map of shadow graphs that is bijective on *in*-arcs at every vertex; such in-coverings induce unital injective $*$ -homomorphisms of graph algebras, but in the direction $\mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$, and on K_0 they induce the *pullback* τ^* . The pushforward τ_* acts by the block matrix $\text{diag}(\tau_*, \tau_*)$ on the Bowen–Franks group $\text{coker}(I - B_k)$, where it is exactly the tail norm of [V]; the two actions are adjoint under the canonical perfect pairing between the torsion of $\text{coker } M$ and of $\text{coker } M^t$. Third, X_∞ is an uncountable profinite group and K -theory commutes with inductive, not projective, limits, so no separable C^* -algebra has $K_0 \supseteq X_\infty$; what the inductive limit $\mathcal{O}_\infty = \varinjlim \mathcal{O}_k$ realizes is the Pontryagin dual. Precisely, $\mathcal{O}_\infty$ is a unital Kirchberg algebra with

$$K_1(\mathcal{O}_\infty) \cong \mathbb{Z}, \quad 0 \rightarrow \mathfrak{X}_\infty^\vee \rightarrow K_0(\mathcal{O}_\infty) \rightarrow \mathbb{Z}[1/q] \rightarrow 0, \quad \mathfrak{X}_\infty := \varprojlim_k (\text{Jac}(Y_k), \tau_*),$$

so that $\mathfrak{X}_\infty \cong \text{Hom}(\text{Tor } K_0(\mathcal{O}_\infty), \mathbb{Q}/\mathbb{Z})$ recovers the pro-module together with its level filtration and the tail norms; by the UCT it also appears in odd K -homology. Along the way the machine found, and we then proved, a structural fact the series had not anticipated: for $q = p$ prime the kernel of the tail norm on $\text{Jac}(Y_{k+1})$ is exactly the p -torsion subgroup, an elementary abelian group of rank $n_k(p - 1) - 1$, so the p -part of the Iwasawa pro-module is a torsion-free compact $\mathbb{Z}_p$ -module of infinite rank, its dual is divisible, and the K_0 sequence above splits. The plain tower algebras $C^*(Y_k)$, by contrast, are all isomorphic to the boundary algebra of [I] and see nothing of the tower.

STATUS OF THE RESULTS

Every numbered statement is proved. The C^* -algebraic inputs are: the Cuntz–Krieger K -theory formula and the gauge-invariant uniqueness theorem for graph algebras [3]; the closure of nuclearity, simplicity, pure infiniteness and the UCT class under inductive limits [1, 4, 5]; the Pontryagin duality between profinite and discrete torsion abelian groups; and Kirchberg–Phillips classification [2], which is quoted once, in Remark 2.6, for an existence statement we do not use. The in-covering lemma (Lemma 2.1) and the dual-graph isomorphism (Proposition 4.1) are proved in full. Every finite identity was machine-verified exactly on the K_4 tower at levels $k \leq 3$ (shadow matrices of size 24, 48, 96; Table 2). Section 5 states what is solved, what remains open, and what is impossible, and records four corrections to the specification that prompted this paper.

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Paper VIII of the series; Papers I–VII are cited as [I]–[VII]. The verification script `tower_kirchberg.py` in the ancillary bundle reproduces every computational claim; its output is archived as `tower_kirchberg_output.txt`.

1. THE LEVEL SHADOW AND ITS NORMALIZATION

Setting as in [V] and [VII]. Y_0 is a finite connected $(q+1)$ -regular graph on n vertices, $q \geq 2$; for $k \geq 1$, Y_k is the Iwahori path graph, a strongly connected q -regular digraph on $n_k = n(q+1)q^{k-1}$ vertices with $Y_{k+1} = L(Y_k)$ the line digraph, so that $V_{Y_{k+1}} = \text{Arcs}(Y_k)$; A_k is the level Hecke (adjacency) operator, $\Delta_k = qI - A_k$ the out-Laplacian, $\text{coker}_{\mathbb{Z}} \Delta_k = \mathbb{Z} \oplus \text{Jac}(Y_k)$ with $\ker \Delta_k = \mathbb{Z}\mathbf{1}$. The degeneracy maps $\tau, \eta: Y_{k+1} \rightarrow Y_k$ are the tail (source) and head (range) maps of arcs; $\tau_*, \eta_* \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ are their pushforwards, $\tau^* = \tau_*^t$, $\eta^* = \eta_*^t$ the pullbacks, and

$$(1.1) \quad \tau_* \eta^* = A_k, \quad \eta^* \tau_* = A_{k+1}, \quad \tau_* \tau^* = \eta_* \eta^* = qI_{n_k}, \quad \tau_* \Delta_{k+1} = \Delta_k \tau_*, \quad \eta_* \Delta_{k+1}^t = \Delta_k^t \eta_*$$

by [V, Prop. 1.2, Thm. 3.1, Prop. 3.2]. Transposing the last two identities,

$$(1.2) \quad \Delta_{k+1}^t \tau^* = \tau^* \Delta_k^t, \quad \Delta_{k+1} \eta^* = \eta^* \Delta_k,$$

so τ^* descends to $\text{coker} \Delta_k^t \rightarrow \text{coker} \Delta_{k+1}^t$ and η^* to $\text{coker} \Delta_k \rightarrow \text{coker} \Delta_{k+1}$, both *up* the tower. We write $\text{Tor}_k := \text{Jac}(Y_k) = \text{Tor coker} \Delta_k$ and $\text{Tor}_k^t := \text{Tor coker} \Delta_k^t$.

Convention 1.1 (Graph algebras). For a finite directed graph E without sinks, $C^*(E)$ is the universal C^* -algebra generated by mutually orthogonal projections p_v ($v \in E^0$) and partial isometries s_e ($e \in E^1$) subject to

$$(CK1) \quad s_e^* s_e = p_{r(e)}, \quad (CK2) \quad p_v = \sum_{s(e)=v} s_e s_e^*,$$

with vertex matrix $A_E(v, w) = \#\{e : s(e) = v, r(e) = w\}$. Then $K_0(C^*(E)) = \text{coker}(1 - A_E^t)$, generated by the classes $[p_v] \mapsto e_v$, and $K_1(C^*(E)) = \ker(1 - A_E^t)$ [3, Thm. 7.16]; this is the convention of [VI], where $K_0(\mathcal{O}_{B(Y)}) = \text{coker}(I - B^t)$. The other standard convention reverses all arcs. Because T_p is symmetric the choice was invisible in [VI]; at the directed levels it is not, and we make it once and for all here. Under the reversed convention every statement of this paper holds with $(\tau, \Delta_k, X_\infty)$ replaced by $(\eta, \Delta_k^t, X_\infty^{\text{in}})$ of [V, Prop. 3.2] (Remark 2.5).

Definition 1.2 (Level shadow). For a strongly connected d -regular digraph Y with adjacency A (every vertex of out-degree $d \geq 2$), the *shadow graph* $Y^\sharp$ is the digraph on $V_Y \sqcup V_Y'$ with the arcs of Y , one arc $v \rightarrow v'$, $c := d - 1$ labelled return arcs $v' \rightarrow v$, and two labelled loops at v' ; its vertex matrix is

$$B(Y) = \begin{pmatrix} A & I \\ cI & 2I \end{pmatrix}, \quad c = d - 1,$$

and the *shadow algebra* is $\mathcal{O}_{B(Y)} := C^*(Y^\sharp)$. For $Y = Y_0$ one has $d = q+1$, $c = q$: this is [VI, Def. 1.1]. For the levels $k \geq 1$ one has $d = q$, $c = q - 1$, and we write $B_k := B(Y_k)$, $\mathcal{O}_k := \mathcal{O}_{B_k}$, $Y_k^\sharp$ for the level- k shadow graph.

Theorem 1.3 (Level K -theory). *Let $k \geq 1$, $c = q - 1$, $U = \begin{pmatrix} I & -cI \\ 0 & I \end{pmatrix}$, $V = \begin{pmatrix} I & 0 \\ -I & I \end{pmatrix}$. Then*

$$(1.3) \quad U(I - B_k^t)V = \Delta_k^t \oplus (-I_{n_k}), \quad V^t(I - B_k)U^t = \Delta_k \oplus (-I_{n_k}),$$

and consequently

$$K_0(\mathcal{O}_k) = \text{coker}(I - B_k^t) \cong \text{coker} \Delta_k^t \cong \mathbb{Z} \oplus \text{Jac}(Y_k), \quad K_1(\mathcal{O}_k) = \ker(I - B_k^t) = \mathbb{Z} \cdot (\mathbf{1}, -\mathbf{1}),$$

the isomorphism $K_0(\mathcal{O}_k) \rightarrow \text{coker} \Delta_k^t$ being $[x] \mapsto [\text{first block of } Ux]$. The Bowen-Franks group $\text{coker}(I - B_k) \cong \text{coker} \Delta_k = \mathbb{Z} \oplus \text{Jac}(Y_k)$ via V^t , and $\ker(I - B_k) = \mathbb{Z} \cdot (\mathbf{1}, -c\mathbf{1})$. The algebra $\mathcal{O}_k$ is a unital Kirchberg algebra (separable, nuclear, simple, purely infinite, UCT), hence traceless. Its unit class is $[1_{\mathcal{O}_k}] = (2 - q) \sum_v [p_v]$; in particular $[1_{\mathcal{O}_k}] = 0$ when $q = 2$.

Proof. $I - B_k^t = \begin{pmatrix} I - A_k^t & -cI \\ -I & -I \end{pmatrix}$, and the two block products of [VI, Thm. 2.1] give $((1+c)I - A_k^t) \oplus (-I)$; with $c = q - 1$ the first block is Δ_k^t . The second identity is the transpose of the first. Cokernels and kernels are invariant under unimodular equivalence, and $\text{coker } \Delta_k^t$ has the same invariant factors as $\text{coker } \Delta_k$, whence the abstract isomorphism with $\mathbb{Z} \oplus \text{Jac}(Y_k)$ (the canonical relation between the two torsion groups is Lemma 1.5). For the kernels: $(x, y) \in \ker(I - B_k^t)$ forces $y = -x$ and $A_k^t x = qx$, and the Perron eigenspace of the irreducible A_k^t (row sums = in-degrees = q) is $\mathbb{Z}\mathbf{1}$, saturated because $\mathbf{1}$ is primitive; $\ker(I - B_k)$ similarly. Kirchberg: $Y_k^\sharp$ is strongly connected (Y_k is, and layers communicate through $v \rightarrow v' \rightarrow v$; here $c \geq 1$ is used), B_k is not a permutation matrix, and every shadow vertex carries two loops, so the criteria quoted in [VI, Thm. 3.1] apply verbatim. Unit class: (CK2) at v' gives $[p_{v'}] = c[p_v] + 2[p_v]$, i.e. $[p_{v'}] = -c[p_v]$, and summing over v gives $[1] = (1 - c) \sum_v [p_v] = (2 - q) \sum_v [p_v]$. $\square$

Remark 1.4 (A correction to the specification: $q - 1$, not q). The specification prompting this paper prescribed $B(Y_k) = \begin{pmatrix} A_k & I \\ qI & 2I \end{pmatrix}$ at every level. The block computation of Theorem 1.3 shows that with c return arcs the Laplacian block is $(1 + c)I - A_k^t$, so the shadow K_0 is the sandpile group only when $1 + c$ equals the out-degree: $c = q$ at the $(q+1)$ -regular base, $c = q - 1$ at the q -regular levels. With $c = q$ at a level $k \geq 1$ the block is $(q+1)I - A_k^t$, nonsingular because every eigenvalue of A_k has modulus $\leq q$; then $K_1 = 0$ and K_0 is a finite group with no relation to $\text{Jac}(Y_k)$. On the K_4 tower it is $\mathbb{Z}/2 \oplus \mathbb{Z}/14 \oplus (\mathbb{Z}/112)^2$ at $k = 1$ and $(\mathbb{Z}/3)^8 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/42 \oplus (\mathbb{Z}/336)^2$ at $k = 2$ (Table 2, line (W)), against $\text{Jac}(Y_1) = \mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/24$ and $\text{Jac}(Y_2) = (\mathbb{Z}/2)^8 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/16 \oplus \mathbb{Z}/48$. The $(q+1)$ -versus- q mismatch is the Eisenstein special layer of the whole series ([I, Rem. 1.8], [V, Rem. 3.3], [VII, Rem. 3.3]); here it fixes the number of return arcs.

Lemma 1.5 (Bowen–Franks duality). *Let $M \in M_n(\mathbb{Z})$. For $[x] \in \text{Tor coker } M$ and $[y] \in \text{Tor coker } M^t$ put*

$$\langle [x], [y] \rangle_M := y^t z \bmod \mathbb{Z}, \quad \text{where } z \in \mathbb{Q}^n, \quad Mz = x.$$

- (i) *The pairing is well defined, bilinear and perfect: $\text{Tor coker } M^t \cong \text{Hom}(\text{Tor coker } M, \mathbb{Q}/\mathbb{Z})$ canonically.*
- (ii) *If $F \in M_{n \times n'}(\mathbb{Z})$ satisfies $FM' = MF$, so that F descends to $\text{coker } M' \rightarrow \text{coker } M$ and F^t to $\text{coker } M^t \rightarrow \text{coker } M'^t$, then $\langle Fx, y \rangle_M = \langle x, F^t y \rangle_{M'}$: the two induced maps are adjoint.*

Proof. (i) If $[x]$ is torsion, $mx = Mz'$ for some $m \neq 0$ and $z = z'/m$ exists; z is unique up to $\ker_{\mathbb{Q}} M$. If $[y]$ is torsion, $my = M^t u$, so for $w \in \ker M$ one has $y^t w = \frac{1}{m} u^t M w = 0$: the ambiguity is invisible. Replacing x by $x + Ma$ changes $y^t z$ by $y^t a \in \mathbb{Z}$; replacing y by $y + M^t b$ changes it by $b^t M z = b^t x \in \mathbb{Z}$. For perfectness write $M = P^{-1} D Q^{-1}$ with D diagonal and P, Q unimodular; then $x \mapsto Px$ identifies $\text{coker } M$ with $\text{coker } D$, $y \mapsto Q^t y$ identifies $\text{coker } M^t$ with $\text{coker } D^t = \text{coker } D$, and if $Mz = x$ then $D(Q^{-1}z) = Px$ and $(Q^t y)^t (Q^{-1}z) = y^t z$, so the pairing is transported to $\langle e_i, e_j \rangle_D = \delta_{ij}/d_i$ on $\bigoplus \mathbb{Z}/d_i$, which is perfect. (ii) If $M'z' = x$ then $M(Fz') = FM'z' = Fx$, so $\langle Fx, y \rangle_M = y^t Fz' = (F^t y)^t z' = \langle x, F^t y \rangle_{M'}$. That F^t descends is $F^t M^t = (MF)^t = (FM')^t = M'^t F^t$. $\square$

Applied to $M = \Delta_k$, $M' = \Delta_{k+1}$, $F = \tau_*$ (using $\tau_* \Delta_{k+1} = \Delta_k \tau_*$), the lemma says: $\text{Tor}_k^t \cong \text{Jac}(Y_k)^\vee := \text{Hom}(\text{Jac}(Y_k), \mathbb{Q}/\mathbb{Z})$ canonically, and under this identification the pullback $\tau^*: \text{Tor}_k^t \rightarrow \text{Tor}_{k+1}^t$ is the Pontryagin dual of the tail norm $\tau_*: \text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)$.

2. THE SHADOW TRANSFER

Lemma 2.1 (In-coverings induce homomorphisms). *Let $p: F \rightarrow E$ be a morphism of finite directed graphs without sinks (maps on vertices and arcs with $s \circ p = p \circ s$, $r \circ p = p \circ r$), surjective on vertices, and an in-covering: for every $w \in F^0$ the restriction $p: r^{-1}(w) \rightarrow r^{-1}(p(w))$ is a bijection.*

Then

$$P_v := \sum_{p(w)=v} p_w, \quad S_e := \sum_{p(f)=e} s_f$$

is a Cuntz–Krieger E -family in $C^*(F)$, and the induced map $p^\sharp: C^*(E) \rightarrow C^*(F)$ is a unital, injective, gauge-equivariant $*$ -homomorphism. Its K_0 -map is the pullback $p^*: \text{coker}(1 - A_E^t) \rightarrow \text{coker}(1 - A_F^t)$, $e_v \mapsto \sum_{p(w)=v} e_w$, which satisfies $p^* A_E^t = A_F^t p^*$; its K_1 -map is the restriction of p^* to $\ker(1 - A_E^t)$.

Proof. Fix $e \in E^1$ with $r(e) = u$. By the in-covering property, r restricts to a bijection from $p^{-1}(e)$ onto $p^{-1}(u)$: each w over u receives exactly one lift of e . Hence for distinct lifts $f \neq f'$ of e one has $r(f) \neq r(f')$ and $s_f s_{f'}^* = s_f p_{r(f)} p_{r(f')}^* s_{f'}^* = 0$, while always $s_f s_{f'} = 0$. Therefore

$$S_e^* S_e = \sum_{p(f)=e} p_{r(f)} = \sum_{p(w)=u} p_w = P_{r(e)}, \quad S_e S_e^* = \sum_{p(f)=e} s_f s_f^*,$$

which is (CK1). For (CK2), using (CK2) in $C^*(F)$ and $s(p(f)) = p(s(f))$,

$$P_v = \sum_{p(w)=v} \sum_{s(f)=w} s_f s_f^* = \sum_{s(p(f))=v} s_f s_f^* = \sum_{s(e)=v} \sum_{p(f)=e} s_f s_f^* = \sum_{s(e)=v} S_e S_e^*.$$

The P_v are mutually orthogonal projections, nonzero by surjectivity, with $\sum_v P_v = \sum_w p_w = 1$. The universal property gives $p^\sharp$; it intertwines the gauge actions since each S_e is a sum of degree-one generators, so the gauge-invariant uniqueness theorem [3, Thm. 2.2] makes it injective. On K_0 , $[P_v] = \sum_{p(w)=v} [p_w]$ is the pullback. Intertwining: $(p^* A_E^t)(w, v) = \#\{\text{arcs } v \rightarrow p(w)\}$ and $(A_F^t p^*)(w, v) = \#\{\text{arcs into } w \text{ from vertices over } v\}$, equal by the bijection on in-arcs. For K_1 let $\mathcal{F}_E = \bigcup_m \mathcal{F}_E^m$ be the gauge-fixed AF core, with $\mathcal{F}_E^m = \text{span}\{s_\mu s_\nu^* : |\mu| = |\nu| = m, r(\mu) = r(\nu)\}$, $K_0(\mathcal{F}_E^m) = \mathbb{Z}^{E^0}$ (the class of $s_\mu s_\mu^*$ being $e_{r(\mu)}$), the inclusions inducing A_E^t , and $K_1(C^*(E))$ identified through the Pimsner–Voiculescu sequence of the dual action with the kernel of $1 - A_E^t$ on $\varinjlim (\mathbb{Z}^{E^0}, A_E^t)$, into which the stage-0 kernel maps isomorphically because A_E^t fixes it [3, Ch. 7]. In-coverings lift paths uniquely backwards from a prescribed terminal vertex: for μ with $r(\mu) = x$ and each w over x there is a unique lift μ_w ending at w , and $p^\sharp(s_\mu) = \sum_w s_{\mu_w}$, so $p^\sharp(s_\mu s_\mu^*) = \sum_w s_{\mu_w} s_{\mu_w}^*$ (cross terms vanish as above), of class $\sum_{p(w)=x} e_w = p^* e_x$ in $K_0(\mathcal{F}_F^m)$. Thus $p^\sharp$ maps $\mathcal{F}_E^m$ into $\mathcal{F}_F^m$ and acts by p^* at every stage, compatibly with A^t by the intertwining, and naturality of the Pimsner–Voiculescu sequence gives $K_1(p^\sharp) = p^*|_{\ker(1 - A_E^t)}$. $\square$

Proposition 2.2 (The tail map is an in-covering of shadows). *For $k \geq 1$ define $\tau^\sharp: Y_{k+1}^\sharp \rightarrow Y_k^\sharp$ by $w \mapsto \tau(w)$, $w' \mapsto \tau(w)'$ on vertices, and on arcs: an arc (e, f) of $Y_{k+1} = L(Y_k)$ (arcs e, f of Y_k with $r(e) = s(f)$) maps to the arc e of Y_k ; the arc $w \rightarrow w'$ maps to $\tau(w) \rightarrow \tau(w)'$; return arcs and loops map to the return arcs and loops of the same label at $\tau(w)'$. Then $\tau^\sharp$ is a surjective graph morphism, bijective on in-arcs at every vertex, and not bijective on out-arcs.*

Proof. Morphism: (e, f) runs from e to f in $L(Y_k)$, and its image e runs from $s(e) = \tau(e)$ to $r(e) = s(f) = \tau(f)$; the shadow arcs are compatible by construction. Surjective since τ is. In-arcs at $f \in V_{Y_{k+1}}$: the q arcs (e, f) with $r(e) = s(f)$, mapping to the q arcs e into $s(f) = \tau(f)$ — all of them, each once — and the c return arcs $f' \rightarrow f$, mapping label-wise onto the return arcs $\tau(f)' \rightarrow \tau(f)$. In-arcs at f' : the arc $f \rightarrow f'$ and the two loops, mapping onto the arc $\tau(f) \rightarrow \tau(f)'$ and the two loops. Out-arcs at f : the q arcs (f, g) with $s(g) = r(f)$ all map to the single arc f of Y_k , so $\tau^\sharp$ is q -to-one there and misses the other out-arcs of $\tau(f)$. $\square$

Theorem 2.3 (Shadow transfer). *For $k \geq 1$ let $\varphi_k := (\tau^\sharp)^\sharp: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$,*

$$\varphi_k(p_v) = \sum_{\tau(w)=v} p_w, \quad \varphi_k(p_{v'}) = \sum_{\tau(w)=v} p_{w'}, \quad \varphi_k(s_a) = \sum_{\tau^\sharp(b)=a} s_b.$$

Then φ_k is a canonical unital injective gauge-equivariant $*$ -homomorphism. Put

$$\mathcal{T}^k := \text{diag}(\tau^*, \tau^*) \in M_{2n_{k+1} \times 2n_k}(\mathbb{Z}).$$

Then:

- (i) $\mathcal{T}^k(I - B_k^t) = (I - B_{k+1}^t)\mathcal{T}^k$, $\mathcal{T}^k \mathbf{1}_{2n_k} = \mathbf{1}_{2n_{k+1}}$, $\mathcal{T}^k(\mathbf{1}, -\mathbf{1}) = (\mathbf{1}, -\mathbf{1})$;
- (ii) $K_0(\varphi_k)$ is the map induced by $\mathcal{T}^k$, which under the isomorphisms of Theorem 1.3 is the pullback $\tau^*: \text{coker } \Delta_k^t \rightarrow \text{coker } \Delta_{k+1}^t$; on torsion it is the Pontryagin dual $\text{Jac}(Y_k)^\vee \rightarrow \text{Jac}(Y_{k+1})^\vee$ of the tail norm τ_* (Lemma 1.5), hence injective; on the torsion-free quotient $\text{coker } \Delta^t / \text{Tor} \cong \mathbb{Z}$ (via the degree) it is multiplication by q ; and $K_0(\varphi_k)$ is injective;
- (iii) $K_1(\varphi_k): \mathbb{Z} \rightarrow \mathbb{Z}$ is the identity.

Proof. Lemma 2.1 applies by Proposition 2.2. (i) The intertwining reduces blockwise to $\tau^* A_k^t = A_{k+1}^t \tau^*$, the transpose of $\tau_* A_{k+1} = A_k \tau_*$ (1.1); $\tau^* \mathbf{1} = \mathbf{1}$ because every vertex of Y_{k+1} has exactly one tail. (ii) Since $\mathcal{T}^k U_k = U_{k+1} \mathcal{T}^k$ (block scalars commute with $\text{diag}(\tau^*, \tau^*)$), the first-block isomorphisms of Theorem 1.3 carry the map induced by $\mathcal{T}^k$ to the map induced by τ^* on $\text{coker } \Delta^t$. The dual of a surjection is injective, and τ_* is surjective on Jac [V, Thm. 3.1(ii)]. Degree: $\deg \circ \Delta^t = (\Delta \mathbf{1})^t = 0$ so $\deg$ descends, and $\deg(\tau^* y) = q \deg y$ because fibres of τ have q elements. If $K_0(\varphi_k)[x] = 0$ then $q \deg x = 0$, so $[x]$ is torsion, so $[x] = 0$. (iii) is Lemma 2.1 with (i). $\square$

Theorem 2.4 (The transfer matrix and the tail norm). *For $k \geq 1$ let $\mathcal{T}_k := \text{diag}(\tau_*, \tau_*) \in M_{2n_k \times 2n_{k+1}}(\mathbb{Z})$. Then:*

- (i) $\mathcal{T}_k(I - B_{k+1}) = (I - B_k)\mathcal{T}_k$, so $\mathcal{T}_k$ induces $(\mathcal{T}_k)_*: \text{coker}(I - B_{k+1}) \rightarrow \text{coker}(I - B_k)$ on the Bowen–Franks groups, and under V^t of (1.3) this map is exactly the tail norm $\tau_*: \text{coker } \Delta_{k+1} \rightarrow \text{coker } \Delta_k$ of [V, Thm. 3.1]: surjective, surjective on Jac , with kernel of order $q^{n_k(q-1)-1}$ on $\text{Jac}(Y_{k+1})$;
- (ii) on $\ker(I - B_{k+1}) = \mathbb{Z}(\mathbf{1}, -c\mathbf{1})$ the matrix $\mathcal{T}_k$ acts by multiplication by $q = \text{ind}(\tau_*)$, the Eisenstein index of [VII, Cor. 2.2];
- (iii) $(\mathcal{T}_k)_*$ on $\text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)$ and $K_0(\varphi_k)$ on $\text{Jac}(Y_k)^\vee \rightarrow \text{Jac}(Y_{k+1})^\vee$ are Pontryagin adjoint;
- (iv) $\mathcal{T}_k$ does not descend to K_0 : $\mathcal{T}_k(I - B_{k+1}^t) - (I - B_k^t)\mathcal{T}_k$ has the single nonzero block $\tau_* A_{k+1}^t - A_k^t \tau_* = \eta_*(qI - \tau^* \tau_*) \neq 0$, whose (v, w) entry is $q[\eta(w) = v] - A_k(\tau(w), v)$.

Proof. (i) Blockwise the intertwining is $\tau_* A_{k+1} = A_k \tau_*$; since $V_k^t \mathcal{T}_k = \mathcal{T}_k V_{k+1}^t$ the induced map on $\text{coker}(\Delta \oplus (-I)) = \text{coker } \Delta$ is the first block τ_* ; the properties listed are [V, Thm. 3.1(ii)]. (ii) $\tau_* \mathbf{1} = q\mathbf{1}$. (iii) is Lemma 1.5(ii) with $F = \tau_*$. (iv) By (1.1), $\tau_* A_{k+1}^t = (A_{k+1} \tau^*)^t = (\eta^* \tau_* \tau^*)^t = q\eta_*$ and $A_k^t \tau_* = (\tau^* A_k)^t = (\tau^* \tau_* \eta^*)^t = \eta_* \tau^* \tau_*$; and $(\eta_* \tau^* \tau_*)(v, w) = \#\{w' : \eta(w') = v, \tau(w') = \tau(w)\} = A_k(\tau(w), v)$, which is 0 or 1 for q different v , whereas $q[\eta(w) = v]$ is q at one v . $\square$

Remark 2.5 (The mirror). Under Convention 1.1 exactly one degeneracy map lifts to the C^* -level: the one that is a covering on the (CK1) side, τ . The head map $\eta^\sharp$ is an out-covering (bijective on out-arcs, the same proof with s and r exchanged), and the sum-over-lifts formula fails for it because the cross terms $s_f s_{f'}$ with $r(f) = r(f')$ do not vanish. Dually, on K_0 the pushforward that descends is η_* ($\eta_* A_{k+1}^t = A_k^t \eta_*$ by (1.1)), not τ_* (Theorem 2.4(iv)). Reversing all arcs exchanges $\tau \leftrightarrow \eta$, $\Delta \leftrightarrow \Delta^t$, and the pro-module X_∞ of [V, Thm. 3.1] with the in-module X_∞^{in} of [V, Prop. 3.2]. This is the C^* -algebraic face of the two-sidedness proved in [V]: the tower is a (τ, η) -bimodule system, and each side is realized by one of the two conventions.

Remark 2.6 (Downward homomorphisms). The specification asked for $*$ -homomorphisms $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ inducing τ_* . Under Convention 1.1 no map $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ can induce τ_* on K_0 , because τ_* does not descend (Theorem 2.4(iv)); the descending pushforward is η_* , and it sends $[1_{\mathcal{O}_{k+1}}] \mapsto q[1_{\mathcal{O}_k}]$ -type classes ($\eta_* \mathbf{1} = q\mathbf{1}$), so any $*$ -homomorphism inducing it is non-unital. By Kirchberg–Phillips classification [2] a $*$ -homomorphism $\psi_k: \mathcal{O}_{k+1} \rightarrow P\mathcal{O}_k P$ with $K_0(\psi_k) = \eta_*$ and $K_1(\psi_k) = q$ exists

for a projection P with $[P] = q[1]$ (quoted, not used); it depends on a choice of KK -lift, ambiguous by $\text{Ext}(K_*(\mathcal{O}_{k+1}), K_{*+1}(\mathcal{O}_k))$, and is not canonical. The canonical object is φ_k ; the tail norm is its K_0 -adjoint, realized on the nose by the integer matrix $\mathcal{T}_k$.

Proposition 2.7 (Prime-to- q stability). *For $k \geq 1$, $\tau_*\eta^* = A_k$ acts on $\text{Jac}(Y_k)$ as multiplication by q , and $\eta^*\tau_* = A_{k+1}$ acts on $\text{Jac}(Y_{k+1})$ as multiplication by q . Hence for every prime $\ell \nmid q$ the tail norm $\tau_*: \text{Jac}(Y_{k+1})[\ell^\infty] \rightarrow \text{Jac}(Y_k)[\ell^\infty]$ is an isomorphism with inverse $q^{-1}\eta^*$, the prime-to- q torsion is the same at every level $k \geq 1$, and $\ker(\tau_*|_{\text{Jac}(Y_{k+1})}) \subseteq \text{Jac}(Y_{k+1})[q]$.*

Proof. η^* descends by (1.2); on $\text{coker } \Delta_k$ one has $A_k \equiv qI$ since $\Delta_k = qI - A_k \equiv 0$; the two composites are (1.1). On the K_4 tower the odd part is $\mathbb{Z}/3$ at $k = 1, 2, 3$ (Table 2). $\square$

Theorem 2.8 (The kernel of the tail norm is the p -torsion). *Let $q = p$ be prime and $k \geq 1$. Then:*

- (i) $\text{rk}_{\mathbb{F}_p} A_{k+1} = n_k$; hence $\dim_{\mathbb{F}_p}(\text{coker } \Delta_{k+1} \otimes \mathbb{F}_p) = n_k(p-1)$ and $\text{Jac}(Y_{k+1})[p] \cong (\mathbb{Z}/p)^{n_k(p-1)-1}$; exactly $n_k(p-1) - 1 = n_{k+1} - n_k - 1$ invariant factors of Δ_{k+1} are divisible by p ;
- (ii) $\ker(\tau_*: \text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)) = \text{Jac}(Y_{k+1})[p]$;
- (iii) the pro-module $\mathfrak{X}_\infty := \varprojlim_k (\text{Jac}(Y_k), \tau_*)$ has no p -torsion. Put $X_\infty^{(p)} := \varprojlim_k (\text{Jac}(Y_k)[p^\infty], \tau_*)$, the torsion part of the module X_∞ of [V, Thm. 3.1]. Then $X_\infty^{(p)}$ is a torsion-free compact $\mathbb{Z}_p$ -module of infinite rank, topologically isomorphic to $\mathbb{Z}_p^{\mathbb{N}}$; its Pontryagin dual is the divisible group $(\mathbb{Q}_p/\mathbb{Z}_p)^{(\mathbb{N})}$; and $\mathfrak{X}_\infty \cong X_\infty^{(p)} \oplus \text{Jac}(Y_1)[p']$, where $\text{Jac}(Y_1)[p']$ is the finite prime-to- p part.

Proof. (i) $A_{k+1} = \eta^*\tau_*$; η^* has full column rank over every field (its columns are indicator vectors of the disjoint nonempty fibres of η) and τ_* is surjective over $\mathbb{Z}$, hence of full row rank n_k over $\mathbb{F}_p$; so $\text{rk}_{\mathbb{F}_p} A_{k+1} = \text{rk}_{\mathbb{F}_p} \tau_* = n_k$. Since $p \mid q$, $\Delta_{k+1} \equiv -A_{k+1} \pmod{p}$, so the $\mathbb{F}_p$ -dimension of $\text{coker } \Delta_{k+1} \otimes \mathbb{F}_p$ is $n_{k+1} - n_k = n_k(p-1)$; one dimension is the free summand $\mathbb{Z}$, the rest is the p -rank of $\text{Jac}(Y_{k+1})$. (ii) The kernel is contained in $\text{Jac}(Y_{k+1})[p]$ by Proposition 2.7; its order is $p^{n_k(p-1)-1}$ by [V, Thm. 3.1(ii)] (the increment of [III, Thm. 2.1], re-proved homologically in [VII, Cor. 2.2]); and $|\text{Jac}(Y_{k+1})[p]| = p^{n_k(p-1)-1}$ by (i). (iii) Let $x = (x_k)_k \in \mathfrak{X}_\infty$ with $px = 0$. For $k \geq 2$, $x_k \in \text{Jac}(Y_k)[p] = \ker \tau_*$ by (ii), so $x_{k-1} = \tau_*x_k = 0$; hence $x = 0$. The p -part is a pro- p abelian group, i.e. a compact $\mathbb{Z}_p$ -module, torsion-free by what was just shown; its Pontryagin dual D is a discrete p -torsion group on which multiplication by p is surjective (dual to injectivity), i.e. divisible, hence $D \cong (\mathbb{Q}_p/\mathbb{Z}_p)^{(I)}$ and $X_\infty^{(p)} \cong \mathbb{Z}_p^I$. Since $X_\infty^{(p)} \twoheadrightarrow \text{Jac}(Y_k)[p^\infty]$ for every k (surjective transitions) and the p -ranks $n_{k-1}(p-1) - 1$ are unbounded, I is infinite; it is countable because a countable inverse limit of finite groups is second countable. The splitting off of the prime-to- p part is Proposition 2.7. Machine confirmation on K_4 ($p = 2$): $\text{rk}_{\mathbb{F}_2} A_2 = 12$, $\text{rk}_{\mathbb{F}_2} A_3 = 24$; the 2-ranks of $\text{Jac}(Y_2)$, $\text{Jac}(Y_3)$ are 11, 23; τ_* annihilates $\text{Jac}(Y_{k+1})[2]$; and $2^{11}, 2^{23}$ are the kernel orders (Table 2, line (Q)). $\square$

The machine found (ii) before we did: at both computed transitions the number of even invariant factors coincided with $\text{ord}_2 |\ker \tau_*|$, and the one-line inclusion of Proposition 2.7 turned the coincidence into a theorem. Its meaning for the Iwasawa-theoretic reading of [III]–[V] is discussed in Remark 4.2. Note that (ii) has no meaning at the boundary transition $0 \rightarrow 1$: there τ_* does not descend to the cokernels ([V, Rem. 3.3]) and $A_1 \neq \eta^*\tau_*$ ([I, Def. 1.7]: $A_1 = T^t S - J$), so the identity $\text{rk}_{\mathbb{F}_p} A_{k+1} = n_k$ driving (i) has no analogue. On K_4 the count of (i) happens to hold at level 1 as well (2-rank 3 = $n_0(q-1) - 1$), but no proof is offered and none is claimed.

3. THE LIMIT ALGEBRA

Definition 3.1. $\mathcal{O}_\infty(Y_\bullet) := \varinjlim_k (\mathcal{O}_k, \varphi_k)$, the C^* -inductive limit of the level shadow algebras along the shadow transfers of Theorem 2.3, starting at $k = 1$. We abbreviate it $\mathcal{O}_\infty$.

Theorem 3.2 (The limit is Kirchberg). $\mathcal{O}_\infty$ is a unital Kirchberg algebra: separable, nuclear, simple, purely infinite, in the UCT class; it admits no trace, finite or semifinite; and it carries the gauge action γ of $\mathbb{T}$ induced from the levels.

Proof. Separability and nuclearity pass to inductive limits. Simplicity: if J is a closed ideal with $J \cap \varphi(\mathcal{O}_k) \neq 0$ for some k then, $\mathcal{O}_k$ being simple and φ injective, $J \ni 1$; otherwise the quotient map is isometric on the dense union $\bigcup_k \varphi(\mathcal{O}_k)$, hence on $\mathcal{O}_\infty$, and $J = 0$. Pure infiniteness passes to inductive limits [1, §4], [4, §4.1], and the bootstrap class is closed under inductive limits [5]. Purely infinite simple algebras have no traces. The φ_k are gauge-equivariant (Lemma 2.1). $\square$

Theorem 3.3 (K -theory of the limit). Write $K_\infty := \varinjlim_k (\text{coker } \Delta_k^t, \tau^*)$ and $\mathfrak{X}_\infty := \varinjlim_k (\text{Jac}(Y_k), \tau_*)$, the full torsion pro-module (its p -part for $p \mid q$ is the torsion of X_∞ of [V, Thm. 3.1]). Then:

- (i) $K_1(\mathcal{O}_\infty) \cong \mathbb{Z}$, generated by the common image of the classes $(\mathbf{1}, -\mathbf{1})_k$;
- (ii) $K_0(\mathcal{O}_\infty) \cong K_\infty$, and each $K_0(\varphi_k)$ is injective, so the levels embed as an exhaustive filtration $K_0(\mathcal{O}_1) \subset K_0(\mathcal{O}_2) \subset \dots \subset K_0(\mathcal{O}_\infty)$ whose k -th torsion term is $\text{Jac}(Y_k)^\vee$, of order $|\text{Jac}(Y_k)|$;
- (iii) $\text{Tor } K_0(\mathcal{O}_\infty) \cong \varinjlim_k (\text{Jac}(Y_k)^\vee, \tau_*^\vee) \cong \text{Hom}_{\text{cont}}(\mathfrak{X}_\infty, \mathbb{Q}/\mathbb{Z}) =: \mathfrak{X}_\infty^\vee$ and $K_0(\mathcal{O}_\infty)/\text{Tor} \cong \mathbb{Z}[1/q]$; thus

$$0 \longrightarrow \mathfrak{X}_\infty^\vee \longrightarrow K_0(\mathcal{O}_\infty) \longrightarrow \mathbb{Z}[1/q] \longrightarrow 0$$

is exact, and $\mathfrak{X}_\infty \cong \text{Hom}(\text{Tor } K_0(\mathcal{O}_\infty), \mathbb{Q}/\mathbb{Z})$ recovers the pro-module, with $\text{Jac}(Y_k) = \text{Hom}(\text{Tor } K_0(\mathcal{O}_k), \mathbb{Q}/\mathbb{Z})$ at each level and τ_* the dual of $K_0(\varphi_k)$;

- (iv) if $q = p$ is prime the sequence splits: $K_0(\mathcal{O}_\infty) \cong \mathbb{Z}[1/p] \oplus (\mathbb{Q}_p/\mathbb{Z}_p)^{(\mathbb{N})} \oplus \text{Jac}(Y_1)[p']$ as an abstract group;
- (v) $[1_{\mathcal{O}_\infty}] = (2 - q)u$ where u is the common image of $\sum_v [p_v]$, a class of infinite order; for $q = 2$ the unit class is 0.

Proof. K -theory commutes with C^* -inductive limits, so $K_*(\mathcal{O}_\infty) = \varinjlim K_*(\mathcal{O}_k)$ along $K_*(\varphi_k)$, computed in Theorem 2.3. (i) The maps are the identity. (ii) Injectivity is Theorem 2.3(ii); the torsion of $K_0(\mathcal{O}_k)$ is $\text{Tor}_k^t \cong \text{Jac}(Y_k)^\vee$ by Lemma 1.5. (iii) Direct limits are exact, so $\text{Tor } \varinjlim = \varinjlim \text{Tor}$ and $\varinjlim (G_k)/\text{Tor} = \varinjlim (G_k/\text{Tor}) = \varinjlim (\mathbb{Z}, \times q) = \mathbb{Z}[1/q]$ (Theorem 2.3(ii)). Pontryagin duality exchanges inverse limits of compact groups with direct limits of their discrete duals, and τ^* is dual to τ_* by Lemma 1.5(ii). The recovery statement is double duality for the discrete torsion group $\mathfrak{X}_\infty^\vee$. (iv) By Theorem 2.8(iii), $\mathfrak{X}_\infty^\vee = D \oplus F$ with D divisible (hence an injective $\mathbb{Z}$ -module) and F finite of order prime to p . $\text{Ext}(\mathbb{Z}[1/p], D) = 0$; and writing $\mathbb{Z}[1/p] = \varinjlim (\mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \dots)$, $\text{Ext}(\mathbb{Z}[1/p], F)$ is the $\varprojlim^1$ of the tower $(F \xleftarrow{p} F \xleftarrow{p} \dots)$, which vanishes because multiplication by p is an automorphism of F . So the extension splits. (v) Theorem 1.3 and $\tau^* \sum_v e_v = \sum_w e_w$; $\deg u = n_k \neq 0$ at level k . $\square$

Corollary 3.4 (The pro-module in K -homology). By the UCT [5] for $\mathcal{O}_\infty$ there are canonical exact sequences

$$0 \rightarrow \text{Ext}(K_0(\mathcal{O}_\infty), \mathbb{Z}) \rightarrow K^1(\mathcal{O}_\infty) \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z} \rightarrow 0,$$

$$0 \rightarrow \left(\prod_{p \mid q} \mathbb{Z}_p \right) / \mathbb{Z} \rightarrow \text{Ext}(K_0(\mathcal{O}_\infty), \mathbb{Z}) \rightarrow \mathfrak{X}_\infty \rightarrow 0.$$

The pro-module itself is therefore a canonical quotient of the Ext-part of the odd K -homology of $\mathcal{O}_\infty$, the kernel being the q -adic Eisenstein term $\text{Ext}(\mathbb{Z}[1/q], \mathbb{Z})$.

Proof. Apply $\text{Hom}(-, \mathbb{Z})$ to the sequence of Theorem 3.3(iii): $\text{Hom}(\mathfrak{X}_\infty^\vee, \mathbb{Z}) = 0$ and $\text{Hom}(\mathbb{Z}[1/q], \mathbb{Z}) = 0$, so $0 \rightarrow \text{Ext}(\mathbb{Z}[1/q], \mathbb{Z}) \rightarrow \text{Ext}(K_0, \mathbb{Z}) \rightarrow \text{Ext}(\mathfrak{X}_\infty^\vee, \mathbb{Z}) \rightarrow 0$. For a torsion group T , $\text{Ext}(T, \mathbb{Z}) \cong \text{Hom}(T, \mathbb{Q}/\mathbb{Z})$ (apply $\text{Hom}(T, -)$ to $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$), so $\text{Ext}(\mathfrak{X}_\infty^\vee, \mathbb{Z}) = \mathfrak{X}_\infty$ by duality. And from $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}[1/q] \rightarrow \bigoplus_{p \mid q} \mathbb{Q}_p/\mathbb{Z}_p \rightarrow 0$ one gets $\text{Ext}(\mathbb{Z}[1/q], \mathbb{Z}) \cong (\prod_{p \mid q} \mathbb{Z}_p)/\mathbb{Z}$, $\mathbb{Z}$ embedded diagonally. $\square$

Theorem 3.5 (No-Go: the pro-module is not a K -group). $\mathfrak{X}_\infty$ (and already its p -part for any $p \mid q$) is an uncountable group, while K_0 and K_1 of a separable C^* -algebra are countable. Hence no separable C^* -algebra — in particular no Kirchberg algebra — has K_0 containing X_∞ or $\mathbb{Z} \oplus X_\infty$, and the formula $K_0(\mathcal{O}_\infty) \cong \mathbb{Z} \oplus X_\infty$ of the specification is false for every candidate $\mathcal{O}_\infty$. The correct statement is the dual one, Theorem 3.3(iii).

Proof. $\mathfrak{X}_\infty \twoheadrightarrow \text{Jac}(Y_k)$ for all k with $|\text{Jac}(Y_k)| \rightarrow \infty$ (the pseudo-Iwasawa law [III, Thm. 2.2]), so $\mathfrak{X}_\infty$ is an infinite profinite group. A countable compact Hausdorff space has an isolated point (Baire); a topological group with an isolated point is discrete; an infinite discrete group is not compact. Hence $\mathfrak{X}_\infty$ is uncountable. Countability of K_* for separable algebras is standard. That K -theory is continuous for inductive and not for projective limits is the reason the inverse system (coker Δ_k, τ_*) cannot be the K_0 of any limit algebra; the inductive system dual to it is. $\square$

4. SPECTRAL BLINDNESS, BROKEN AT THE TOWER LEVEL

Proposition 4.1 (The plain tower algebras are constant). *For $k \geq 1$ the tail map $\tau: Y_{k+1} = L(Y_k) \rightarrow Y_k$ is an in-covering of plain graphs, and $\tau^\natural: C^*(Y_k) \rightarrow C^*(Y_{k+1})$ is an isomorphism, with inverse $\Psi: p_e \mapsto s_e s_e^*, s_{(e,f)} \mapsto s_e s_f s_f^*$. Consequently $C^*(Y_k) \cong C^*(Y_1) = \mathcal{O}_{A_1}$ for all $k \geq 1$, $\varinjlim_k C^*(Y_k) \cong \mathcal{O}_{A_1}$, and at every level and in the limit*

$$K_0 \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r-1), \quad K_1 \cong \mathbb{Z}^r, \quad r = \frac{(q-1)n}{2} + 1,$$

the boundary-channel groups of [I, Thm. 3.7]; the rooted tail groupoid algebra of [I, Prop. 1.9(iii)] and [III, §1] has $K_0 \cong \mathbb{Z}/(q-1)$, $K_1 = 0$. Neither sees $\text{Jac}(Y_k)$ for any k .

Proof. In-arcs at $f \in L(Y_k)^0$ are the (e, f) with $r(e) = s(f)$, mapping bijectively to the in-arcs e of $\tau(f) = s(f)$: Lemma 2.1 gives $\tau^\natural$. The family $\{s_e s_e^*, s_e s_f s_f^*\}$ is a Cuntz–Krieger $L(Y_k)$ -family in $C^*(Y_k)$: $(s_e s_f s_f^*)^*(s_e s_f s_f^*) = s_f s_f^* p_{r(e)} s_f s_f^* = s_f s_f^*$ since $p_{s(f)} s_f = s_f$, and $\sum_{s(f)=r(e)} s_e s_f s_f^* s_f s_f^* s_e^* = s_e p_{r(e)} s_e^* = s_e s_e^*$; so Ψ exists. $\Psi \tau^\natural(p_v) = \sum_{s(e)=v} s_e s_e^* = p_v$, $\Psi \tau^\natural(s_e) = \sum_{s(f)=r(e)} s_e s_f s_f^* = s_e$; $\tau^\natural \Psi(p_e) = \sum_{f, f'} s_{(e,f)} s_{(e,f')}^* = \sum_f s_{(e,f)} s_{(e,f)}^* = p_e$ (cross terms vanish, $r(e, f) = f \neq f'$), and $\tau^\natural \Psi(s_{(e,f)}) = (\sum_{f_1} s_{(e,f_1)}) p_f = s_{(e,f)}$. The K -groups are Morita invariants of $C^*(Y_1) = \mathcal{O}_{A_{\text{nb}}}$, Morita equivalent to $C(\partial X) \rtimes \Gamma$ [I, Prop. 1.9(ii)]. On K_4 : $\text{coker}(I - A_k^t) = \mathbb{Z}^3 \oplus \mathbb{Z}/2$ at $k = 1, 2, 3$, $r = 3$ (Table 2, line (B)). $\square$

algebra	K_0	K_1	tower content
rooted tail groupoid $C_r^*(\mathcal{G}_\infty)$ [I, III]	$\mathbb{Z}/(q-1)$	0	none
plain tower limit $\varinjlim C^*(Y_k) \cong \mathcal{O}_{A_1}$	$\mathbb{Z}^r \oplus \mathbb{Z}/(r-1)$	$\mathbb{Z}^r$	none: constant in k
shadow limit $\mathcal{O}_\infty(Y_\bullet)$	$0 \rightarrow \mathfrak{X}_\infty^\vee \rightarrow K_0 \rightarrow \mathbb{Z}[1/q] \rightarrow 0$	$\mathbb{Z}$	all of it, dually

TABLE 1. Spectral blindness and its breaking. The shadow limit carries the Pontryagin dual of the full torsion pro-module, filtered by levels; $\text{Jac}(Y_k)$ is recovered at level k as $\text{Hom}(\text{Tor } K_0(\mathcal{O}_k), \mathbb{Q}/\mathbb{Z})$ and the tail norm as the dual of $K_0(\varphi_k)$.

Remark 4.2 (Where μ , $-k$ and the Euler line live). The pseudo-Iwasawa law $\text{ord}_p |\text{Jac}(Y_k)| = \mu p^k - k + \nu_0$, $\mu = n(q+1)/q$ [III, Thm. 2.2], is the growth law of the torsion of the level filtration of $K_0(\mathcal{O}_\infty)$ (Theorem 3.3(ii): 7, 18, 41 on K_4). The Eisenstein index $\text{ind}(\tau_*) = q$ of [VII] appears three times in the limit: as the action of $\mathcal{T}_k$ on $\ker(I - B)$ (Theorem 2.4(ii)), as the scaling of the free part of K_0 that produces $\mathbb{Z}[1/q]$ (Theorem 2.3(ii)), and as the q -adic term $\text{Ext}(\mathbb{Z}[1/q], \mathbb{Z})$ separating K -homology from the pro-module (Corollary 3.4). Theorem 2.8 sharpens the arithmetic

reading: for $q = p$ the p -rank of $\text{Jac}(Y_{k+1})$ is $(n_{k+1} - n_k) - 1$, the arc increment minus the Eisenstein line, exactly the two terms of the snake formula [VII, eq. (2.1)] now read as a rank count rather than an order count; and the p -part of the pro-module is $\mathbb{Z}_p^{\mathbb{N}}$ — torsion-free of infinite $\mathbb{Z}_p$ -rank. In classical Iwasawa theory the exponential term μp^k in the growth of the finite layers is produced by the p -power-torsion part of a finitely generated $\mathbb{Z}_p[[T]]$ -module, which is infinite whenever $\mu > 0$; here the same growth is produced by a module with no p -torsion at all. This is a third certificate, alongside [III, Prop. 2.3] ($\lambda = -1$) and [V, Prop. 4.1] ($A_k A_k^t \neq A_k^t A_k$), that the classical mechanism is not the one at work. We state it as a certificate, not as a theory: Problem 5.1 of [V] and its residue [VII, Rem. 3.1] are untouched by this paper.

5. SCOPE, RESIDUE, CORRECTIONS

Remark 5.1 (Solved, open, impossible). *Solved*: the tower-level functoriality of [VI, Rem. 4.3]. The tail norms lift canonically to the shadow algebras — as unital injective $*$ -homomorphisms $\varphi_k: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$ whose K_0 -maps are the Pontryagin adjoints of τ_* , with τ_* itself realized by the integer transfer matrix $\mathcal{T}_k$ on the Bowen–Franks side — and the lifts assemble into a unital Kirchberg algebra $\mathcal{O}_\infty$ with $K_1 \cong \mathbb{Z}$ and $\text{Tor } K_0 \cong \mathfrak{X}_\infty^\vee$, from which the pro-module, its level filtration and the tail norms are recovered by duality. *Open*: (a) a canonical C^* -correspondence $\mathcal{O}_{k+1} \rightsquigarrow \mathcal{O}_k$ induced by the out-covering $\eta^\sharp$ with K_0 -map η_* ; existence is abstract (Remark 2.6), canonicity is not established; (b) whether the sequence of Theorem 3.3(iii) splits for composite q ; (c) the Hecke-module structure of $\mathfrak{X}_\infty$ as a module over the degeneracy pro-ring of [V, Prop. 4.1], and with it [V, Prob. 5.1]. *Impossible*: a separable C^* -algebra with $K_0 \supseteq X_\infty$ (Theorem 3.5); and, as before, a Hecke-dynamical realization ([II, Thm. 1.6], [VI, Rem. 4.3]).

Remark 5.2 (Four corrections to the task specification). (1) The level shadow needs $q - 1$ return arcs, not q (Remark 1.4). (2) The matrix $\mathcal{T}_k = \text{diag}(\tau_*, \tau_*)$ does not induce a map $\text{coker}(I - B_{k+1}^t) \rightarrow \text{coker}(I - B_k^t)$ (Theorem 2.4(iv)); it induces τ_* on $\text{coker}(I - B_{k+1}) \rightarrow \text{coker}(I - B_k)$, the Bowen–Franks groups, which are the Pontryagin duals of the K_0 -groups on torsion. The K_0 -map of the canonical $*$ -homomorphism is the pullback τ^* , in the opposite direction. (3) The target $K_0(\mathcal{O}_\infty) \cong \mathbb{Z} \oplus X_\infty$ is impossible (Theorem 3.5); the theorem is $\text{Tor } K_0(\mathcal{O}_\infty) \cong \mathfrak{X}_\infty^\vee$, $K_0/\text{Tor} \cong \mathbb{Z}[1/q]$, $K_1 \cong \mathbb{Z}$. (4) The blind algebra with $K_0 \cong \mathbb{Z}/(q - 1)$ is the rooted tail groupoid algebra of [I, Prop. 1.9(iii)] and [III, §1]; [I, Thm. 3.7] is the boundary channel, with $K_0 \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r - 1)$; both are blind, and the plain tower algebras are the latter at every level (Proposition 4.1). We also decline the specification’s historical superlative; what is proved is stated in Remark 5.1.

Remark 5.3 (Boundary level). The transition $0 \rightarrow 1$ is not covered: $\tau: Y_1 \rightarrow Y_0$ is not an in-covering (in-degrees q against $q+1$), the shadow normalizations differ ($c = q$ at level 0, $c = q - 1$ at level 1), the intertwining fails by the degeneracy boundary [V, Rem. 3.3], and $A_1 \neq \eta^* \tau_*$. As everywhere in the series the base enters as the initial value ν_0 , and the limit $\mathcal{O}_\infty$ starts at $k = 1$. The single-level algebra of [VI] at level 0 stands beside the tower, not at its foot.

6. THE VERIFICATION BATTERY

Table 2 lists the exact checks performed by `tower_kirchberg.py`; the letters match the docstring of the script and the references in the text. The levels $k \leq 3$ of the K_4 tower are the same ones used in [V] and [VII], so every number here can be compared line by line with the batteries of those papers.

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Unrefereed preprint. Every numbered statement is proved; the C^* -algebraic inputs are quoted with flagged citations in the Status paragraph, and Kirchberg–Phillips classification is quoted only for an existence statement that nothing depends on (Remark 2.6). Theorem 2.8 was found by

check	levels	result
(S) both factorizations (1.3), $c = q - 1$	$k \leq 3$	exact
(K) $\text{SNF}(I - B_k^t)$: one zero; torsion = $\text{Jac}(Y_k)$; $\text{ord}_2 = 7, 18, 41$; odd part $\mathbb{Z}/3$	$k \leq 3$	exact
(W) $c = q$: no zero invariant, finite $K_0 \neq \mathbb{Z} \oplus \text{Jac}$	$k \leq 3$	Rem. 1.4
(C) $\tau^\sharp$: morphism, surjective, bijective on in-arcs, not on out-arcs	$1 \rightarrow 2, 2 \rightarrow 3$	exact
(U) $\mathcal{T}^k(I - B_k^t) = (I - B_{k+1}^t)\mathcal{T}^k$; $\mathcal{T}^k \mathbf{1} = \mathbf{1}$; $\mathcal{T}^k(\mathbf{1}, -\mathbf{1}) = (\mathbf{1}, -\mathbf{1})$	$1 \rightarrow 2, 2 \rightarrow 3$	exact
(D) $\mathcal{T}_k(I - B_{k+1}) = (I - B_k)\mathcal{T}_k$; transposed version fails; $\mathcal{T}_k(\mathbf{1}, -c\mathbf{1}) = q(\mathbf{1}, -c\mathbf{1})$	$1 \rightarrow 2, 2 \rightarrow 3$	exact
(N) $(\mathcal{T}_k)_*$ on torsion: surjective, $ \ker = 2^{11}, 2^{23}$; identical to τ_* on Jac	$2 \rightarrow 1, 3 \rightarrow 2$	exact
(Q) $\text{rk}_{\mathbb{F}_2} A_{k+1} = n_k$; 2-rank of $\text{Jac} = n_k - 1$; τ_* kills $\text{Jac}[2]$; $ \text{Jac}[2] = \ker \tau_* $	$2 \rightarrow 1, 3 \rightarrow 2$	exact
(P) $\langle \tau_* x, y \rangle_k = \langle x, \tau^* y \rangle_{k+1}$ in $\mathbb{Q}/\mathbb{Z}$ on random torsion classes	$1 \rightarrow 2, 2 \rightarrow 3$	exact
(G) strong connectivity, double loop at a shadow vertex	$k \leq 3$	exact
(B) $\text{SNF}(I - A_k^t) = \mathbb{Z}^3 \oplus \mathbb{Z}/2$ at every level	$k \leq 3$	exact
(L) $\deg \tau^* y = q \deg y$, $\deg \tau_* x = \deg x$	$1 \rightarrow 2, 2 \rightarrow 3$	exact

TABLE 2. Exact integer verification (`tower_kirchberg.py`) on the K_4 tower, $q = 2$, shadow matrices of size 24, 48, 96. Invariant factors found: $\text{Jac}(Y_1) = \mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/24$, $\text{Jac}(Y_2) = (\mathbb{Z}/2)^8 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/16 \oplus \mathbb{Z}/48$, $\text{Jac}(Y_3) = (\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^8 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/32 \oplus \mathbb{Z}/96$. For the fixed tower each line is itself a proof; the general statements are proved in the text.

machine and then proved; it is stated for $q = p$ prime, and for composite q only the inclusion $\ker \tau_* \subseteq \text{Jac}[q]$ is claimed. Remark 5.2 records the corrections to the specification that prompted the paper. Bibliographic data of [3, 2, 4, 5] were checked against publisher or indexing pages (titles, venues, years, volumes); page ranges are as commonly cited.

REFERENCES

- [1] E. Kirchberg and M. Rørdam, *Non-simple purely infinite C^* -algebras*, Amer. J. Math. **122** (2000), 637–666.
- [2] N. C. Phillips, *A classification theorem for nuclear purely infinite simple C^* -algebras*, Doc. Math. **5** (2000), 49–114.
- [3] I. Raeburn, *Graph algebras*, CBMS Regional Conference Series in Mathematics **103**, Amer. Math. Soc., Providence, RI, 2005.
- [4] M. Rørdam, *Classification of nuclear, simple C^* -algebras*, in: Classification of nuclear C^* -algebras. Entropy in operator algebras, Encyclopaedia Math. Sci. **126**, Springer, Berlin, 2002, pp. 1–145.
- [5] J. Rosenberg and C. Schochet, *The Künneth theorem and the universal coefficient theorem for Kasparov’s generalized K -functor*, Duke Math. J. **55** (1987), 431–474.

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